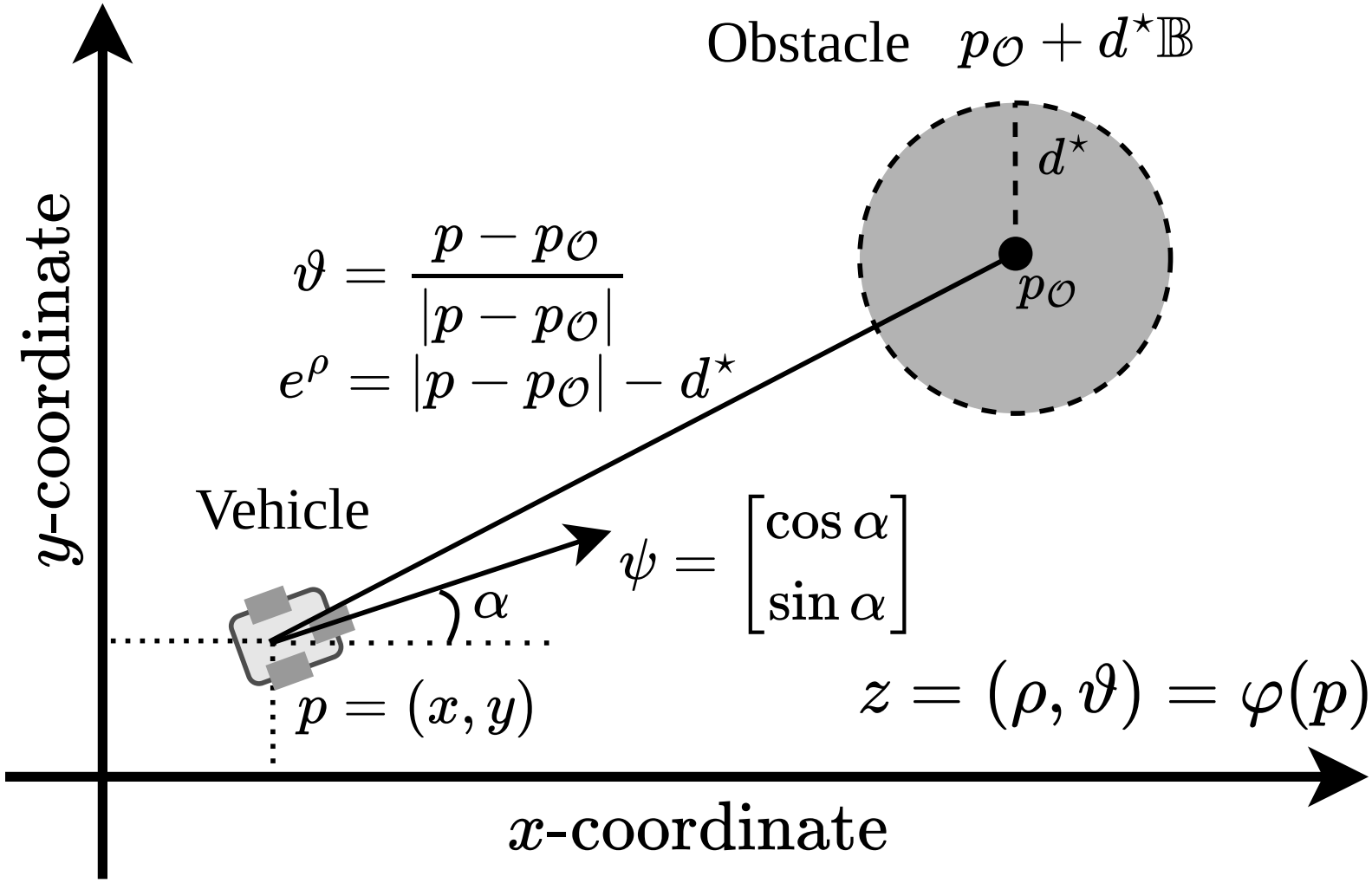
Obstacle $p_{\mathcal{O}} + d^\star \mathbb{B}$
$d^\star$
$p_{\mathcal{O}}$
$\vartheta = \dfrac{p - p_{\mathcal{O}}}{|p - p_{\mathcal{O}}|}$
$e^\rho = |p - p_{\mathcal{O}}| - d^\star$
Vehicle
$\psi = \begin{bmatrix} \cos\alpha \\ \sin\alpha \end{bmatrix}$
$\alpha$
$p = (x, y)$
$z = (\rho, \vartheta) = \varphi(p)$
$y$-coordinate
$x$-coordinate